

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First Semester of M. Pharm (Drug Regulatory Affairs)

University Theory Examination January 2018

PHRA001 PHARMACEUTICAL FORMULATION DEVELOPMENT

Date: 05.01.18 , Friday

Time: 10:00 a.m.to1:00p.m.

Maximum Marks: 70

Instructions:

1. Q.1 and Q.4 are compulsory.
2. Figures to right indicate marks.
3. Section wise separate answer book to be used.
4. Draw neat sketches wherever necessary.

SECTION-I

- Q.1 (A) Explain the importance of carrier drug delivery system. 05
(B) Write methods of preparation for liposome. 05
Q.2 A) Discuss novel controlled parenteral drug delivery system. 05
B) Write a note on monoclonal antibodies. 05

OR

- Q.2 A) Write a note on Transdermal Drug Delivery System. 05
(B) Discuss in details on SMEDDS. 05
Q.3 Answer the following (Any three) 15
(A) Explain mucoadhesive controlled drug delivery system.
(B) Describe formulation aspect and evaluation of effervescent tablet.
(C) Write a note on nano emulsion.
(D) Describe formulation aspect and evaluation of oral film.

SECTION-II

- Q.4 (A) Write a shortnote on IIG limit. 05
(B) Explain in brief how excipient – package interaction takes place? 05
Q.5 Write a short note on 'Accelerated stability study'. 10

OR

- Q.5 How dissolution media can be selected? Explain in detail with proper justification. 10
Q.6 Answer the following (Any three) 15
(A) List out the chemical factors affecting on formulation of drugs.
(B) Explain the role of surface area on formulation of drugs.
(C) Write a shortnote on drug's solubility's importance in pharmaceutical formulation.
(D) Explain in brief about automation in dissolution testing methods.